

Empirical Methods in Finance

Class 7 - Non-Stationarity and Cointegration

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Objectives of the lecture

Some financial variables are naturally modeled in returns, but many economically important variables are not stationary in levels: prices, exchange rates, interest rates, or dividends.

This lecture has two goals:

1. understand how to detect non-stationarity,
2. understand when non-stationary series can still move together in the long run.

Objectives of the lecture

We will therefore study:

1. non-stationarity and unit roots,
2. testing non-stationarity,
3. cointegration,
4. testing cointegration,
5. VECM dynamics and the Johansen approach,
6. an illustration with the dividend discount model.

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Non-stationarity

A stationary series fluctuates around a stable benchmark. Many financial variables do not.

Typical examples are:

- log stock prices,
- exchange rates,
- short- and long-term interest rates,
- dividends or earnings,
- spot and futures prices for the same asset.

Why this matters:

- standard regressions in levels can be misleading,
- correlations can look strong even when there is no economic relationship,
- if a long-run equilibrium exists, we need tools that separate *short-run moves* from *long-run adjustment*.

The most classical non-stationary process is the **random walk**:

$$\begin{aligned} p_t &= p_{t-1} + \varepsilon_t \\ &= p_0 + \sum_{i=1}^t \varepsilon_i, \end{aligned}$$

where p_0 is the starting value and $\{\varepsilon_t\}$ is a white-noise process.

Intuition.

- Every shock is permanently accumulated in the level of the series.
- There is no force pushing the process back to a fixed mean.
- The best forecast is simply the current value:

$$\hat{p}_h(k) = \mathbb{E}[p_{h+k} \mid p_h, p_{h-1}, \dots] = p_h, \quad k > 0.$$

Therefore, a random walk is **not mean reverting**.

A more realistic specification for log prices is the **random walk with drift**:

$$p_t = \mu + p_{t-1} + \varepsilon_t,$$

where $\mu = \mathbb{E}[p_t - p_{t-1}]$ is the average growth rate.

Its moving-average representation is

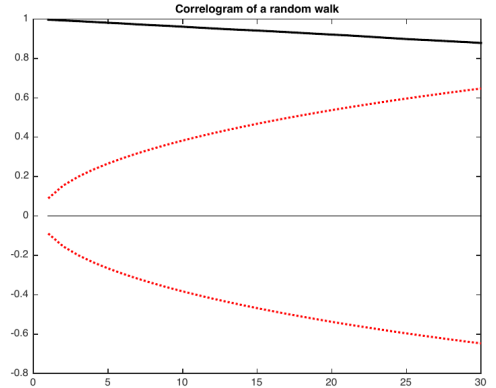
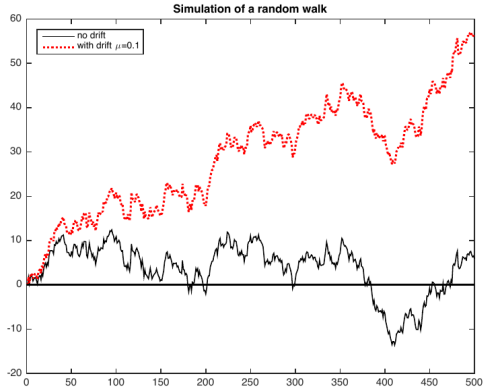
$$p_t = t\mu + p_0 + \sum_{i=1}^t \varepsilon_i.$$

Interpretation.

- $t\mu$ generates a deterministic trend,
- $\sum_{i=1}^t \varepsilon_i$ accumulates all past shocks,
- shocks still have a permanent effect on the series.

This is exactly why prices can trend for long periods without being stationary.

Random Walk: visual intuition



For the random walk with drift, the first moments depend on calendar time:

$$\mathbb{E}[p_t] = p_0 + t\mu,$$

$$\text{Var}(p_t) = t\sigma_\varepsilon^2,$$

$$\text{Cov}(p_t, p_{t-k}) = (t-k)\sigma_\varepsilon^2.$$

Hence

$$\text{Corr}(p_t, p_{t-k}) = \sqrt{\frac{t-k}{t}} \rightarrow 1 \quad \text{as } t \rightarrow \infty \text{ for fixed } k.$$

Key lesson.

- The mean and variance are not time invariant.
- This violates weak stationarity.
- High autocorrelation in levels is therefore not evidence of a meaningful economic mechanism.

The random walk with drift can be written as

$$\Delta p_t = \mu + \varepsilon_t.$$

So **one difference** is enough to recover a stationary process.

More generally, if a series must be differenced d times to become stationary and the differenced series follows an $\text{ARMA}(p, q)$ model, then the original series follows an $\text{ARIMA}(p, d, q)$ model:

$$\phi(L)\Delta^d p_t = \mu + \theta(L)\varepsilon_t.$$

We then say that p_t is integrated of order d , denoted $I(d)$.

Testing non-stationarity

Testing logic: what are we really testing?

Most unit-root tests compare two worlds:

- **Null hypothesis:** the series contains a unit root and is non-stationary,
- **Alternative hypothesis:** the series is stationary around a constant or a deterministic trend.

A convenient benchmark is the AR(1) model:

$$p_t = \varphi_1 p_{t-1} + \varepsilon_t.$$

Interpretation of φ_1 .

- if $|\varphi_1| < 1$, shocks die out and the series is stationary,
- if $\varphi_1 = 1$, shocks never die out and we have a random walk.

So the key question becomes: **is φ_1 statistically equal to 1?**

Dickey-Fuller test: case with no constant

Assume

$$p_t = \varphi_1 p_{t-1} + \varepsilon_t,$$
$$\varepsilon_t \sim iid N(0, \sigma_\varepsilon^2), \quad p_0 = 0.$$

We test

$$H_0 : \varphi_1 = 1 \quad (\text{random walk without drift}),$$
$$H_a : \varphi_1 < 1 \quad (\text{stationary AR}(1)).$$

Dickey-Fuller test: case with no constant

The OLS estimator is

$$\hat{\varphi}_1 = \frac{\sum_{t=1}^T p_{t-1} p_t}{\sum_{t=1}^T p_{t-1}^2},$$

so that

$$\hat{\varphi}_1 - \varphi_1 = \frac{\sum_{t=1}^T p_{t-1} \varepsilon_t}{\sum_{t=1}^T p_{t-1}^2}.$$

Why the Dickey-Fuller distribution is non-standard

If $|\varphi_1| < 1$, then

$$\sqrt{T}(\hat{\varphi}_1 - \varphi_1) \overset{a}{\sim} N\left(0, \frac{\sigma_\varepsilon^2}{\mathbb{E}[p_{t-1}^2]}\right) = N(0, 1 - \varphi_1^2),$$

because

$$\mathbb{E}[p_{t-1}^2] = \frac{\sigma_\varepsilon^2}{1 - \varphi_1^2}.$$

But under the null $H_0 : \varphi_1 = 1$, the usual $\sqrt{T}$ asymptotics collapse.

Why the Dickey-Fuller distribution is non-standard

We then need a different scaling:

$$T(\hat{\varphi}_1 - 1) = \frac{T^{-1} \sum_{t=1}^T p_{t-1} \varepsilon_t}{T^{-2} \sum_{t=1}^T p_{t-1}^2}.$$

Implication: the t -statistic does not follow a standard normal distribution under the unit-root null.

Dickey-Fuller test: practical consequence

Two important consequences follow.

Consequence 1. $\hat{\varphi}_1$ is **super-consistent** when $\varphi_1 = 1$.

Consequence 2. The Dickey-Fuller statistic

$$\tau = \frac{\hat{\varphi}_1 - 1}{\text{std}(\hat{\varphi}_1)} = \frac{\hat{\varphi}_1 - 1}{\left(\hat{\sigma}_\varepsilon^2 / \sum_{t=1}^T p_{t-1}^2\right)^{1/2}},$$

with

$$\hat{\sigma}_\varepsilon^2 = \frac{1}{T-1} \sum_{t=1}^T (p_t - \hat{\varphi}_1 p_{t-1})^2,$$

must be compared to **special Dickey-Fuller critical values**, not to standard Gaussian ones.

Dickey-Fuller test: constant and trend specifications

The deterministic terms matter.

Case 1: constant term

$$p_t = \mu + \varphi_1 p_{t-1} + \varepsilon_t.$$

The associated statistic is usually denoted τ_μ .

Case 2: constant and linear trend

$$p_t = \mu + \gamma t + \varphi_1 p_{t-1} + \varepsilon_t.$$

The associated statistic is usually denoted τ_τ .

Takeaway.

- Different deterministic components imply different null distributions.
- In practice, do not mix the critical values from the three cases:
 1. no constant,
 2. constant,
 3. constant plus trend.

Augmented Dickey-Fuller (ADF) test

The basic Dickey-Fuller test assumes white-noise innovations. If residuals are serially correlated, the critical values are no longer valid.

The ADF idea is simple: add lags of Δp_t to absorb serial correlation:

$$p_t = \mu + \varphi_1 p_{t-1} + \sum_{i=1}^{k-1} \delta_i \Delta p_{t-i} + \varepsilon_t.$$

The null remains

$$H_0 : \varphi_1 = 1.$$

Lag selection.

- Said and Dickey show that k may increase slowly with sample size,
- a common practical rule is $k \approx [T^{1/4}]$,
- a very reasonable classroom rule is: add lags until residual autocorrelation disappears.

The KPSS test complements ADF by reversing the null hypothesis.

It is based on the representation

$$y_t = \beta^\top D_t + \mu_t + \eta_t,$$

$$\mu_t = \mu_{t-1} + \epsilon_t,$$

where D_t contains deterministic components.

Null hypothesis: the series is stationary.

Alternative: the series contains a random-walk component.

The test statistic is

$$KPSS = T^{-2} \sum_{t=1}^T \frac{\hat{S}_t^2}{\hat{\lambda}^2},$$

where $\hat{S}_t = \sum_{j=1}^t \hat{\nu}_j$ and $\hat{\lambda}^2$ is a long-run variance estimator.

Pedagogical rule. ADF asks “is there a unit root?” while KPSS asks “can I safely regard this series as stationary?”

A simple decision rule for class

Using ADF and KPSS together is often more informative than using only one test.

- **ADF rejects + KPSS does not reject:** strong evidence in favor of stationarity.
- **ADF does not reject + KPSS rejects:** strong evidence in favor of non-stationarity.
- **Both reject:** probably misspecified deterministic terms, structural break, or strong persistence.
- **Neither rejects:** the sample is not very informative; be cautious.

For students:

- always state the null hypothesis,
- always state whether you include a drift or a trend,
- always explain what the result means economically.

Cointegration

Spurious regression

Regressing one non-stationary variable on another often creates a **spurious regression**.

Consider two independent random walks:

$$y_t = y_{t-1} + v_t,$$

$$x_t = x_{t-1} + w_t.$$

If we run

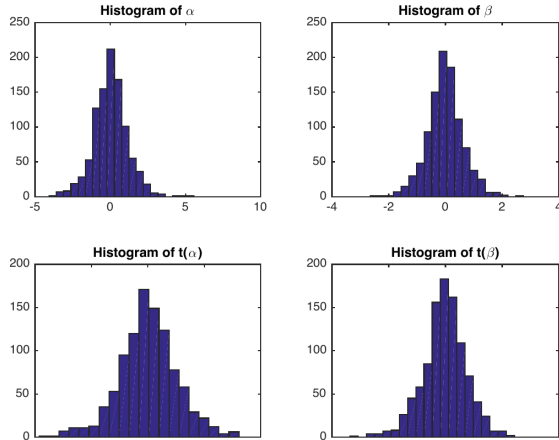
$$y_t = \hat{\alpha}_T + \hat{\beta}_T x_t + \hat{u}_t,$$

we often obtain:

- large R^2 ,
- apparently significant t -statistics,
- highly persistent residuals,
- and yet **no true economic link**.

This is why “high fit in levels” is not enough when variables are trending.

Spurious regression: simulation evidence



Suppose y_t and x_t are both $I(1)$. In general, any linear combination is also $I(1)$:

$$u_t = y_t - \alpha - \beta x_t.$$

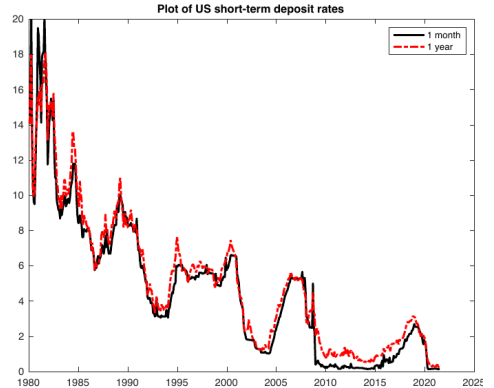
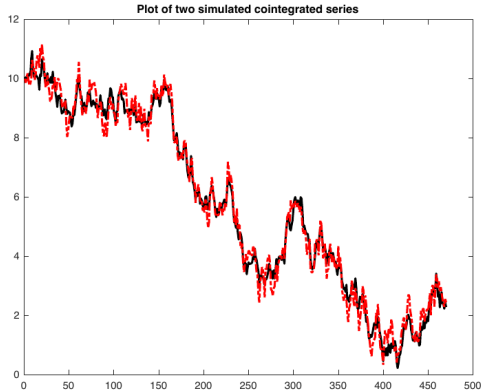
But if u_t turns out to be $I(0)$, then y_t and x_t are **cointegrated**.

Economic meaning.

- The two series may drift in the short run,
- but they cannot drift apart forever,
- because a long-run equilibrium ties them together.

This stationary residual u_t is exactly the **equilibrium error**.

Cointegration: visual intuition



Three ways to explain cointegration in class

1. **Long-run equilibrium.** Two prices may wander, but economic forces prevent them from diverging indefinitely.
2. **Common stochastic trend.** Each series is non-stationary, but they share the same dominant trend component.
3. **Error-correction mechanism.** When the spread gets too large, future changes tend to pull the variables back toward equilibrium.

This third interpretation is the bridge to the **Vector Error Correcting Model (VECM)**.

Testing cointegration

Engle-Granger two-step approach

The simplest route to cointegration is a two-step procedure.

Step 1: estimate the long-run relation

$$y_t = \alpha + \beta x_t + u_t.$$

Step 2: test whether the residual is stationary

$$\hat{u}_t = y_t - \hat{\alpha}_T - \hat{\beta}_T x_t.$$

If $\hat{u}_t$ is stationary, then the two variables are cointegrated.

Good classroom intuition:

- first estimate the equilibrium line,
- then test whether deviations from that line are temporary or permanent.

Residual-based tests: Durbin-Watson and its limits

A classical residual-based statistic is the Durbin-Watson statistic:

$$dw = \frac{\sum_{t=2}^T (\hat{u}_t - \hat{u}_{t-1})^2}{\sum_{t=1}^T \hat{u}_t^2}.$$

Under the null of **no cointegration**, the residual is $I(1)$, so dw tends to be close to 0.

However, the DW approach is limited:

- the asymptotic distribution is non-standard,
- nuisance parameters matter,
- the critical bounds depend on the number of regressors,
- higher-order autocorrelation is awkward to handle.

So in practice we usually prefer an ADF-style residual test.

Engle and Granger propose a residual-based ADF regression:

$$\Delta \hat{u}_t = (\phi - 1)\hat{u}_{t-1} + \sum_{i=1}^p \theta_i \Delta \hat{u}_{t-i} + v_t.$$

We test whether the residual still contains a unit root.

Important subtlety.

- $\hat{u}_t$ is itself estimated from a first-step regression,
- therefore the critical values are **not** the standard Dickey-Fuller ones,
- they depend on the number of regressors in the cointegration equation.

This is the core logic behind the Phillips-Ouliaris and Engle-Granger residual tests.

When Engle-Granger is not enough

Engle-Granger is very useful, but it has three major limits:

1. it is designed mainly for **one** cointegration relation,
2. results can depend on which variable is normalized on the left-hand side,
3. it does not naturally handle systems with several jointly endogenous variables.

This is exactly why the **Johansen approach** is so important in multivariate finance.

It treats cointegration as a **system property**, not as a by-product of one single regression.

Johansen approach

A low-level introduction to the Johansen idea

Start from an n -dimensional VAR(p) in levels:

$$X_t = \mu + A_1 X_{t-1} + \cdots + A_p X_{t-p} + \varepsilon_t.$$

If the components of X_t are $I(1)$, we rewrite the VAR in differences:

$$\Delta X_t = \mu + \Pi X_{t-1} + \sum_{i=1}^{p-1} \Gamma_i \Delta X_{t-i} + \varepsilon_t.$$

This is already a **VECM**. The whole Johansen method is about understanding the matrix Π .

Why? Because Π contains the long-run information.

The key decomposition: $\Pi = \alpha\beta^\top$

If the rank of Π is r with $0 < r < n$, then we can write

$$\Pi = \alpha\beta^\top,$$

where

- β contains the r cointegration vectors,
- $\beta^\top X_{t-1}$ are the long-run equilibrium errors,
- α contains the adjustment speeds.

Interpretation.

- β tells us *what equilibrium means*,
- α tells us *which variables react when equilibrium is violated*.

This is the multivariate version of residual-based error correction.

How to read the rank in Johansen's method

The rank of Π has a direct economic meaning:

$$\text{rank}(\Pi) = r.$$

- $r = 0$: no cointegration, so the system is a VAR in first differences.
- $0 < r < n$: there are r stationary long-run relations among the variables.
- $r = n$: the vector X_t is stationary in levels.

Example. With three $I(1)$ variables:

- $r = 1$ means one equilibrium relation,
- $r = 2$ means two independent equilibrium relations,
- $r = 0$ means no common long-run restriction at all.

Johansen tests for the number of cointegration relations

Johansen proposes likelihood-ratio tests based on the ordered eigenvalues

$$\hat{\lambda}_1 > \dots > \hat{\lambda}_n.$$

Trace test

$$LR_{\text{trace}}(r) = -T \sum_{i=r+1}^n \log(1 - \hat{\lambda}_i).$$

Maximum eigenvalue test

$$LR_{\text{max}}(r, r+1) = -T \log(1 - \hat{\lambda}_{r+1}).$$

In words.

- the trace test asks whether the number of cointegration relations is at most r ,
- the max-eigenvalue test asks whether the rank is exactly r against $r+1$.

For an empirical exercise, the steps are:

1. check that the variables are individually $I(1)$,
2. choose the VAR lag order in levels,
3. decide whether to include a constant and/or deterministic trend,
4. run Johansen rank tests,
5. estimate β (long-run relations),
6. interpret α (adjustment speeds),
7. then analyze short-run dynamics in the VECM.

Pedagogical summary. Johansen answers two questions at once:

- *How many* equilibrium relations are there?
- *Who adjusts* when the system is away from equilibrium?

VECM dynamics

Suppose two variables satisfy the long-run relation

$$y_t = \alpha + \beta x_t + z_t,$$

where z_t is stationary.

Then the relevant short-run dynamics is not a VAR in levels and not a pure VAR in differences. It is the **Vector Error Correcting Model**:

$$\Delta x_t = \mu_1 + \phi_{11}\Delta x_{t-1} + \phi_{12}\Delta y_{t-1} + \gamma_1 z_{t-1} + \varepsilon_{1,t},$$

$$\Delta y_t = \mu_2 + \phi_{21}\Delta x_{t-1} + \phi_{22}\Delta y_{t-1} + \gamma_2 z_{t-1} + \varepsilon_{2,t}.$$

The new term z_{t-1} is the **error-correction term**.

How to read a VECM economically

Suppose y_{t-1} is too high relative to the long-run equilibrium, so that $z_{t-1} > 0$.

Then we typically expect:

- Δy_t to fall, which suggests $\gamma_2 < 0$,
- Δx_t to rise, which suggests $\gamma_1 > 0$.

In words:

- a positive equilibrium error means the spread is too large,
- the coefficients on z_{t-1} measure how the system closes that gap,
- the larger $|\gamma_i|$, the faster variable i adjusts.

If both error-correction coefficients are close to zero, there is little evidence of meaningful equilibrium adjustment.

Engle-Granger and Johansen in one picture

Engle-Granger

- estimate one long-run equation,
- test residual stationarity,
- then estimate a VECM if cointegration is accepted.

Johansen

- starts from the full VAR system,
- estimates the number of cointegration relations directly,
- delivers β and α simultaneously.

A good rule for class.

- with two variables and one clear equilibrium story, Engle-Granger is often enough,
- with several variables or several long-run relations, use Johansen.

Illustration: DDM

Illustration: the Dividend Discount Model

Theory. Let the holding return satisfy

$$R_{t+1} = \frac{P_{t+1} + D_{t+1} - P_t}{P_t},$$

where P_t is the stock price and D_t the dividend.

If expected returns are constant, $\mathbb{E}_t[R_{t+1}] = R$, then the fundamental price obeys

$$P_t = \frac{1}{1+R} \mathbb{E}_t[P_{t+1} + D_{t+1}].$$

Let $\delta = \frac{1}{1+R}$. Using iterated expectations, we obtain

$$P_t = \sum_{i=1}^{T-1} \delta^i \mathbb{E}_t[D_{t+i}] + \delta^T \mathbb{E}_t[P_{t+T} + D_{t+T}].$$

Illustration: long-run price-dividend relation

If the stock is held indefinitely and the transversality condition holds,

$$\lim_{T \rightarrow \infty} \delta^T \mathbb{E}_t[P_{t+T} + D_{t+T}] = 0,$$

then

$$P_t = \sum_{i=1}^{\infty} \frac{\mathbb{E}_t[D_{t+i}]}{(1+R)^i}.$$

Assume dividends follow

$$D_{t+1} = (1 + \mu)D_t + \varepsilon_{t+1}.$$

The DDM therefore transforms a valuation formula into a potential long-run restriction between price and dividend.

Illustration: from dividend dynamics to cointegration

Under the AR(1) dividend growth assumption,

$$\mathbb{E}_t[D_{t+i}] = (1 + \mu)^i D_t,$$

so the fundamental value becomes

$$P_t^* = D_t \frac{1 + \mu}{r - \mu}.$$

Taking logs, we obtain

$$p_t = a + b d_t + u_t,$$

$$u_t = p_t - p_t^*,$$

with

$$a = \log \left(\frac{1 + \mu}{r - \mu} \right), \quad b = 1.$$

If the DDM is right, u_t should be stationary.

ADF test with drift:

$$p_t = \mu + \varphi_1 p_{t-1} + \sum_{i=1}^{k-1} \delta_i \Delta p_{t-i} + \varepsilon_t.$$

	U.S. Price	U.S. Dividend	U.K. Price	U.K. Dividend
$t(\varphi_1)$	-1.9284	-2.4888	-1.9962	-2.5423
p-value	0.6418	0.3377	0.6053	0.3114

Conclusion. All four series behave as non-stationary processes in levels.

Empirical illustration: estimated cointegration relation

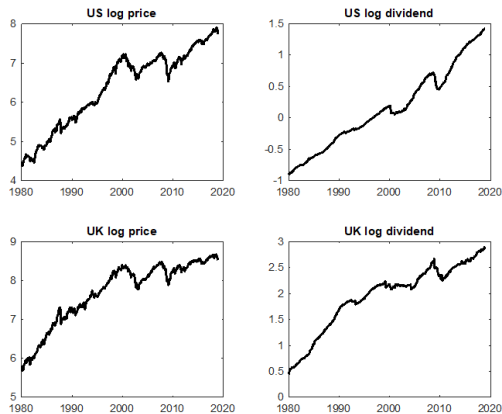
Cointegration regression:

$$p_t = a + bd_t + u_t.$$

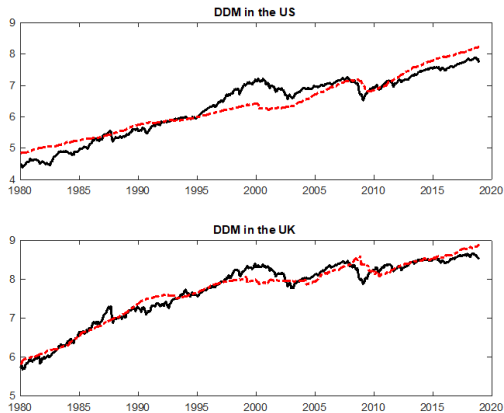
	U.S.	U.K.
$\hat{a}$	6.0899	5.3186
t -value	416.11	242.58
$\hat{b}$	1.3919	1.2219
t -value	83.35	106.62
R^2	0.9217	0.9495

A high R^2 by itself does *not* prove cointegration. The key test is whether the residual u_t is stationary.

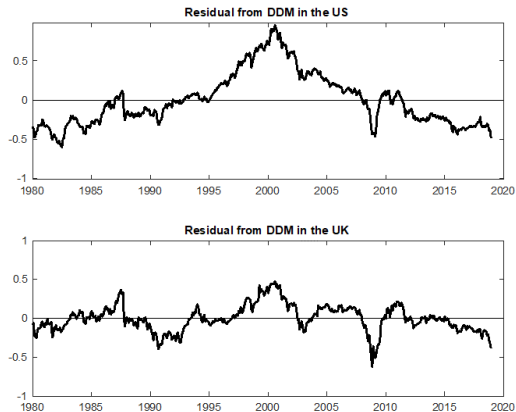
Empirical illustration: the data



Empirical illustration: fitted equilibrium relation



Empirical illustration: residuals from the DDM relation



Empirical illustration: residual-based cointegration test

ADF test with no constant term:

$$u_t = \varphi_1 u_{t-1} + \sum_{i=1}^{k-1} \delta_i \Delta u_{t-i} + \varepsilon_t.$$

	U.S.	U.K.
$t(\hat{\phi} - 1)$	-1.8429	-3.4680
p-value	0.063	0.000

Interpretation.

- U.S. residuals are stationary at the 10% level,
- U.K. residuals are stationary at the 1% level,
- the long-run price-dividend relation is therefore much stronger in the U.K. sample.

Empirical illustration: VECM for the U.S.

Using the VECM introduced earlier, the U.S. estimates are:

U.S. (no cointegration)	Parameter	<i>t</i> -stat
μ_1	0.0081	3.1831
ϕ_{11}	0.0614	1.0181
ϕ_{12}	-0.1332	-0.6245
γ_1	-0.0125	-1.7186
μ_2	0.0042	7.1163
ϕ_{21}	-0.0005	-0.0361
ϕ_{22}	0.1542	2.9196
γ_2	0.0007	0.3148

The error-correction terms are weak and economically unconvincing, which is consistent with the weaker cointegration evidence in the U.S.

Empirical illustration: VECM for the U.K.

Using the VECM introduced earlier, the U.K. estimates are:

U.K. (cointegration)	Parameter	<i>t</i> -stat
μ_1	0.0056	2.5857
ϕ_{11}	0.0360	0.6102
ϕ_{12}	-0.0230	-0.2315
γ_1	-0.0223	-1.6703
μ_2	0.0041	3.7841
ϕ_{21}	0.0361	1.0398
ϕ_{22}	-0.0133	-0.1777
γ_2	0.0169	1.7447

The signs of γ_1 and γ_2 fit the error-correction story much better in the U.K. data.

What you should remember

1. Non-stationary series can create convincing but false regressions.
2. Cointegration means that a particular long-run spread is stationary.
3. Engle-Granger is the simplest way to test one equilibrium relation.
4. Johansen generalizes the idea to a full multivariate system.
5. A VECM separates **short-run dynamics** from **long-run equilibrium adjustment**.

Cointegration lets us model short-run fluctuations without giving up the idea of a meaningful long-run equilibrium.

Application

What is done in this notebook

- Load quarterly data on US GDP, the S&P 500, US industrial production and oil prices.
- Work in logs and check visually whether the variables look persistent.
- Run ADF and KPSS tests in levels and in first differences.
- Estimate two regressions in the spirit of Engle and Granger:
 - a deliberately spurious benchmark,
 - an economically more defensible candidate relation.
- Run a first Johansen cointegration exercise on a trivariate system.
- Estimate a rank-1 VECM and interpret the error-correction mechanism.

Variables and transformation

We use the four variables below:

- US GDP,
- S&P 500,
- US industrial production,
- oil prices.

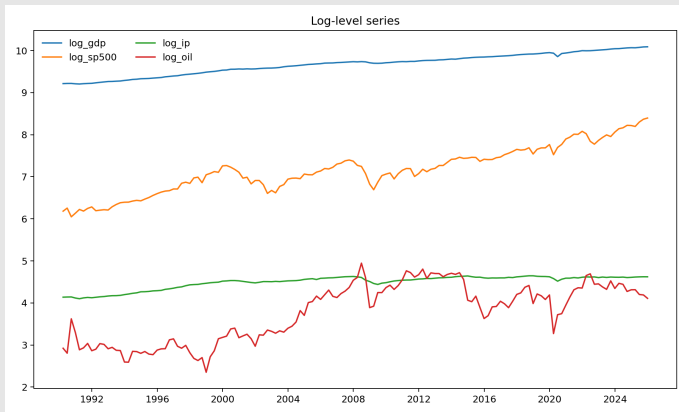
The notebook works with log-levels:

$$x_t = \log(X_t) \quad (1)$$

This is convenient because:

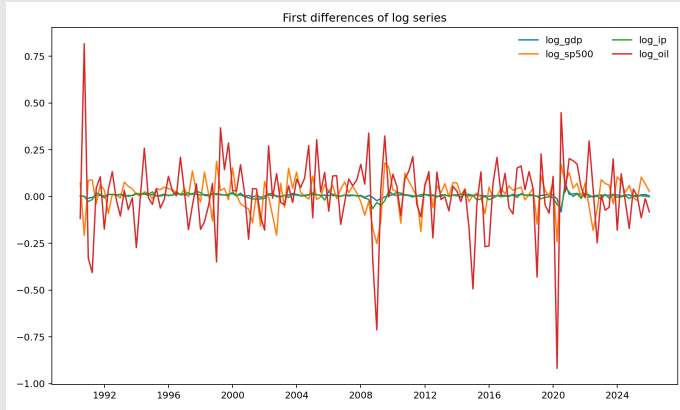
- trends become easier to compare,
- first differences of logs are close to growth rates,
- long-run relations are easier to interpret.

Visual inspection



The four log-level series are highly persistent and display clear low-frequency movements.

Visual inspection after differencing



Once differenced, the variables fluctuate around much more stable means.

Stationarity tests

Series	ADF p(level)	ADF p(diff)	KPSS p(level)	KPSS p(diff)
log_gdp	0.762	0.000	0.010	0.100
log_sp500	0.413	0.000	0.054	0.100
log_ip	0.749	0.000	0.010	0.100
log_oil	0.248	0.000	0.010	0.100

Interpretation:

- in levels, ADF does not reject a unit root;
- in levels, KPSS often rejects stationarity;
- in first differences, the evidence points to stationarity.

The dataset is therefore consistent with variables that are broadly $I(1)$.

A first regression that looks strong but is spurious

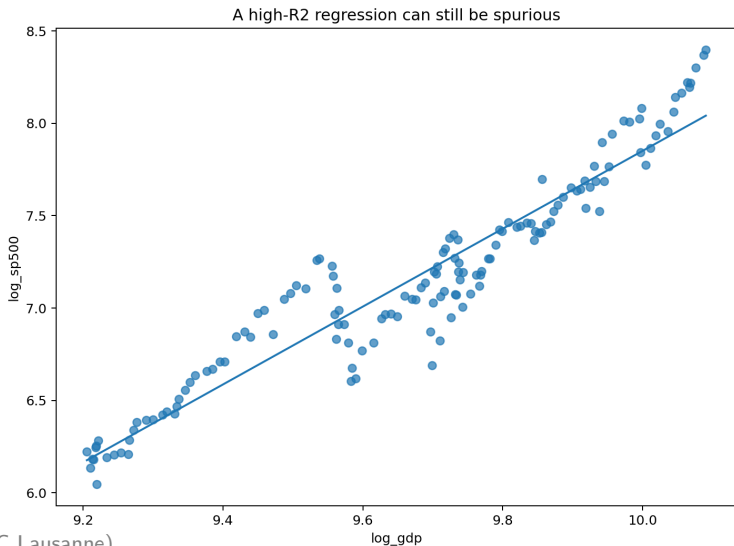
We estimate:

$$\log(SP500)_t = \alpha + \beta \log(GDP)_t + u_t \quad (2)$$

Main output:

- $R^2 = 0.905$,
- Durbin-Watson = 0.229,
- Engle-Granger tau statistic = -2.571 ,
- p-value = 0.249.

A first regression that looks strong but is spurious



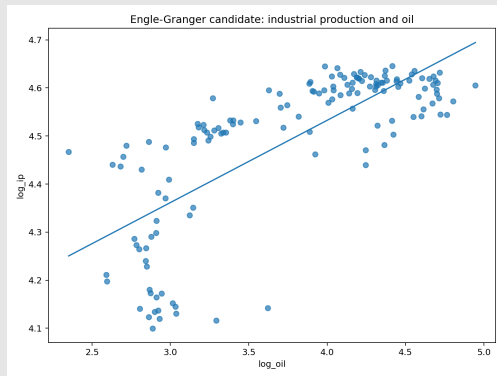
We next estimate:

$$\log(IP)_t = \alpha + \beta \log(Oil)_t + u_t \quad (3)$$

Main output:

- $R^2 = 0.559$,
- Durbin-Watson = 0.108,
- Engle-Granger tau statistic = -2.685 ,
- p-value = 0.205.

Engle-Granger candidate relation



This pair is more plausible economically, but the residual-based test still does not reject no cointegration at 5%.

Why move to Johansen?

Engle-Granger has two important limits:

- it depends on one normalisation only;
- it can have low power in finite samples.

Johansen starts from a system:

$$\Delta y_t = \Pi y_{t-1} + \Gamma_1 \Delta y_{t-1} + \cdots + \Gamma_p \Delta y_{t-p} + \varepsilon_t \quad (4)$$

and factorises:

$$\Pi = \alpha \beta' \quad (5)$$

The rank of Π tells us how many long-run stationary relations exist inside the system.

Johansen application: trivariate system

We test the system ($\log GDP$, $\log SP500$, $\log IP$) with 5 lagged differences and no deterministic term inside the cointegration relation.

Null	Trace stat	5% cv	Reject?
$r = 0$	31.666	29.796	Yes
$r \leq 1$	13.443	15.494	No
$r \leq 2$	1.948	3.842	No

Null	Max-eigen stat	5% cv	Reject?
$r = 0$	18.222	21.131	No
$r = 1$	11.495	14.264	No
$r = 2$	1.948	3.842	No

Reading: the trace test points to rank 1, while the max-eigen test is more conservative.

VECM with one cointegration relation

We estimate a rank-1 VECM and normalise the cointegration vector on industrial production:

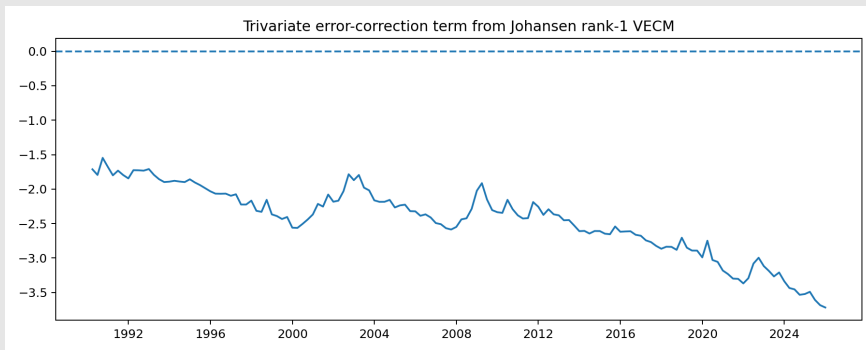
$$ECT_{t-1} = 0.163 \log(GDP)_{t-1} - 1.189 \log(SP500)_{t-1} + \log(IP)_{t-1} \quad (6)$$

Loading coefficients:	Equation	α	p-value
	$\Delta \log(GDP)_t$	-0.0006	0.000
	$\Delta \log(SP500)_t$	-0.0021	0.045
	$\Delta \log(IP)_t$	-0.0002	0.187

Interpretation:

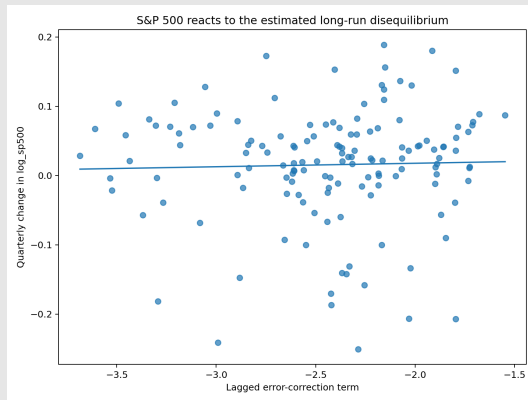
- GDP and the stock market help absorb disequilibrium;
- industrial production reacts less clearly in this specification.

Error-correction term



The ECT_{t-1} measures the distance from the estimated long-run equilibrium.

Illustration of adjustment



When the lagged disequilibrium is high, the subsequent change in the S&P 500 tends to be weaker. That is the intuition behind the error-correction mechanism.

Takeaways for students

- Step 1: check integration orders before running regressions in levels.
- Step 2: remember that a high R^2 can be fully misleading with non-stationary data.
- Step 3: use Engle-Granger as a simple first diagnostic.
- Step 4: use Johansen when the problem is genuinely multivariate.
- Step 5: once a rank is chosen, the VECM separates:
 - the long-run relation $\beta' y_{t-1}$,
 - the short-run dynamics,
 - the speeds of adjustment in α .

Pedagogically, the key message is:

Unit roots $\rightarrow$ cointegration $\rightarrow$ VECM (7)